Qualcomm Technologies, Inc.

Adding U.FL Antenna Connectors to DragonBoard™ 410c and Validating GPS on Android and Linux

Application Note

LM80-P0436-42 Rev. C

January 18, 2018

Questions or comments: https://www.96boards.org/DragonBoard410c/forum

Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

Revision history

Revision	Date	Description
A	October 2015	Initial release
B	September 2016	Update to 'E' part
C	January 2018	Added Section 3.6 Updated Table Table 3-2

Contents

Figures

Tables

1 Introduction

1.1 Purpose

This application note provides instructions on how to add U.FL connectors (for connecting external Wi-Fi or GPS antenna) to the DragonBoard 410c development platform. These instructions are required if the DragonBoard 410c is mounted inside a conductive enclosure. In such a scenario, the on-board Wi-Fi and GPS antenna cannot pick up Wi-Fi or GPS signals from outside the enclosure. Therefore, external Wi-Fi and GPS antennas are required, which are mounted outside the conductive enclosure and connected to the U.FL connectors.

NOTE: These instructions are valid for DragonBoard 410c boards that have shields populated on the processor, power management chip, and memory. Older DragonBoard 410c boards without shields do not function properly for GPS.

1.2 Acronyms, abbreviations, and terms

Table 1-1 provides definitions for the acronyms, abbreviations, and terms used in this document.

Table 1-1 Acronyms, abbreviations, and terms

Term	Definition
GPS	Global Positioning System
SNR	Signal to Noise Ratio
U.FL	U.FL is a miniature RF connector for high-frequency signals up to 6 GHz manufactured by Hirose Electric Group and others.

2 Wi-Fi U.FL connector

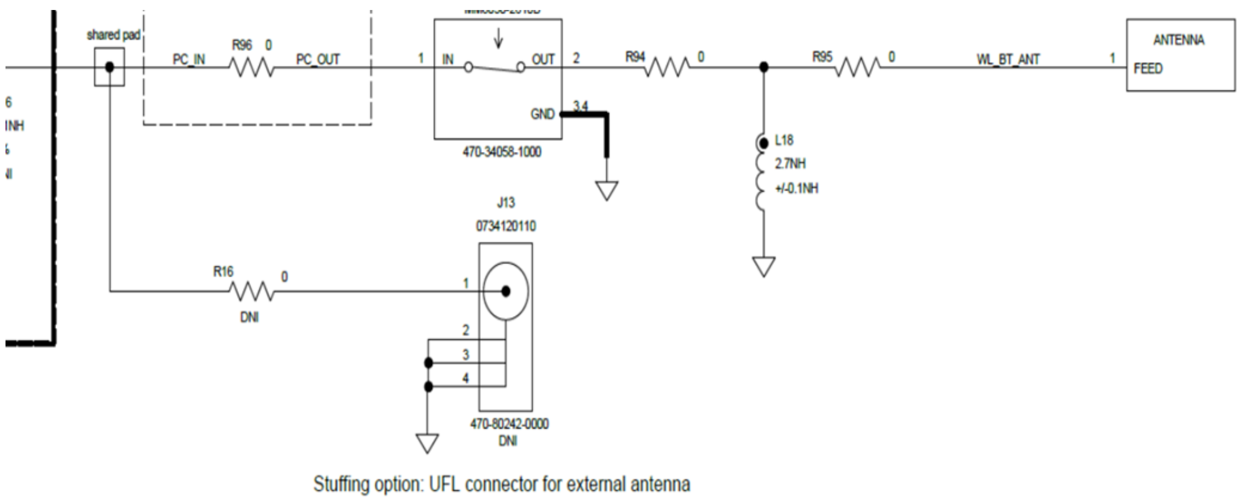

Figure 2-1 DragonBoard 410c schematic showing Wi-Fi antenna circuitry

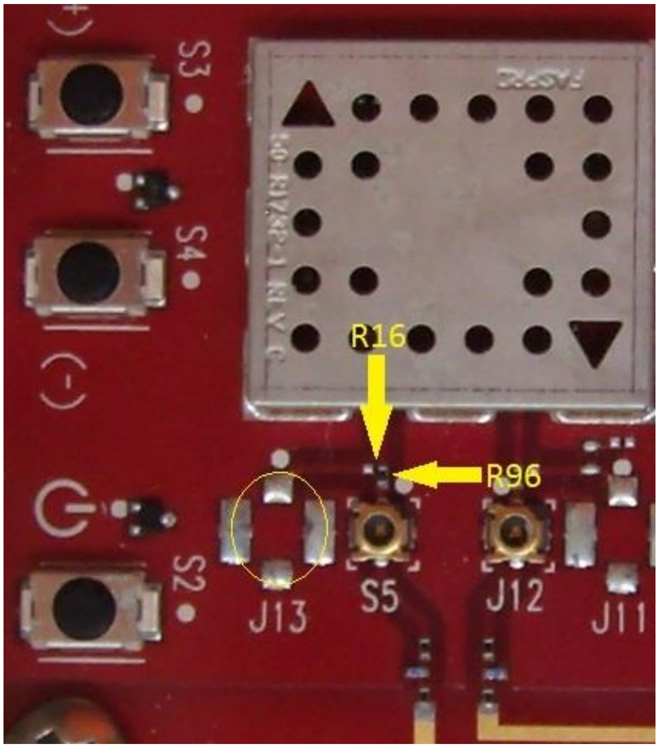

Figure 2-2 Locations of components on circuit board for adding Wi-Fi U.FL connector

Table 2-1 Components required for Wi-Fi U.FL connector on DragonBoard 410c

Description	Location (on board)	Vendor	Vendor Part No.	DigiKey Part No.	Mouser Part No.
U.FL connector, 1.25 mm surface mount, 50 Ohm	J13	Molex	Molex 73412-0110	WM5587CT-ND	538-73412-0110
External Wi-Fi Antenna	Connect to U.FL connector on J13	Multiple	Taoglas FXP73.07.0100A	931-1076-ND	960-FXP73070100A

2.1 Procedure to connect W-Fi U.FL connector

To connect the Wi-Fi U.FL connector, perform the following:

1. Solder on J13 (the U.FL connector) .
2. Move R96 to R16.

NOTE: The resistor is of zero-Ohm. While doing the rework, it is easier to remove R96 and bridge R16 with solder rather than trying to solder on one of the small resistors.

3 GPS U.FL Connector

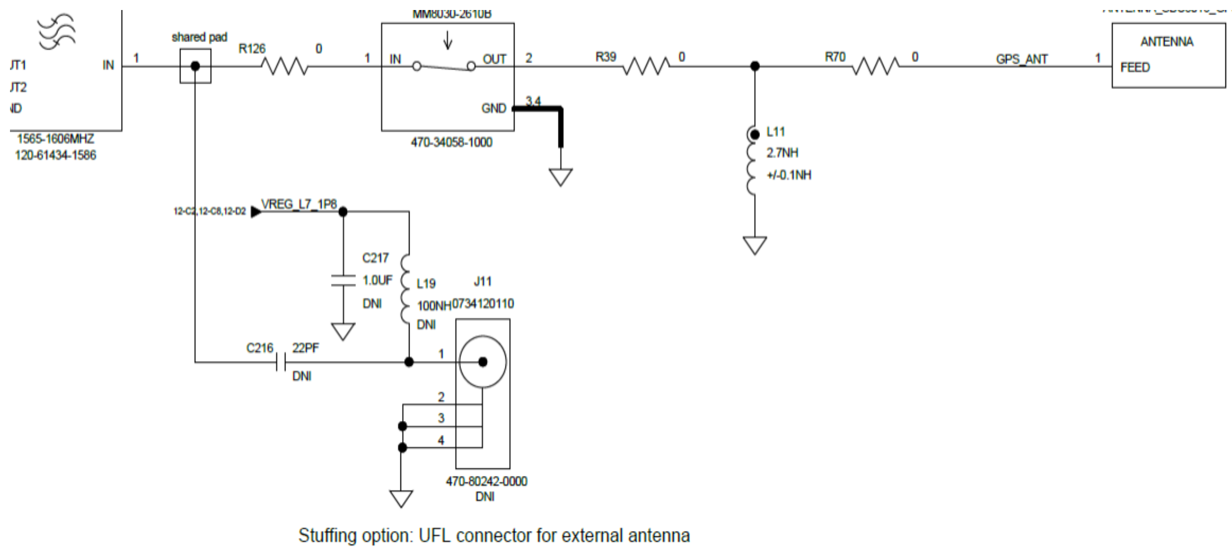

Figure 3-1 DragonBoard 410c schematic showing GPS antenna circuitry

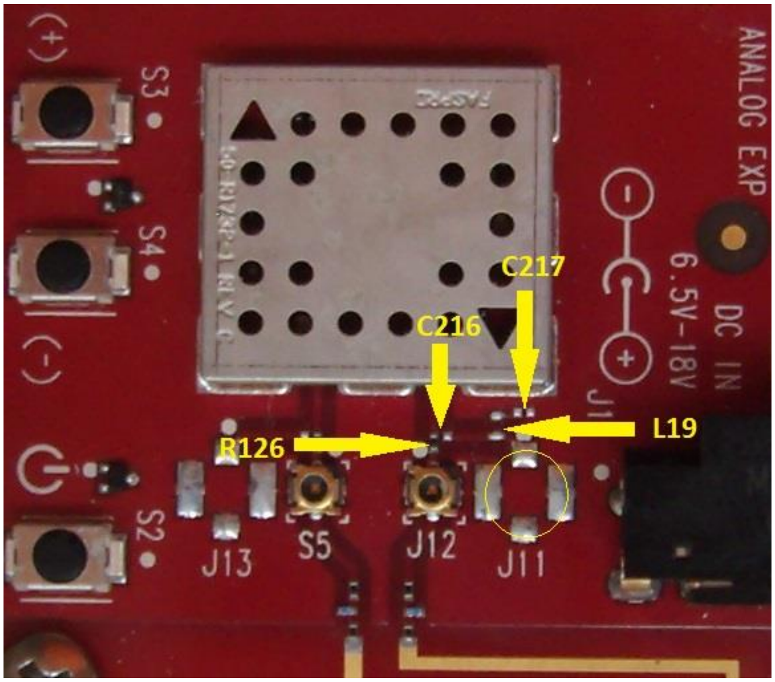

Figure 3-2 Locations of components on circuit board for adding GPS U.FL connector

The options for using an external GPS antenna (with rework instructions) are as follows:

- Option 1: Passive GPS antenna - Requires a minimal rework on the board, but provides lower GPS sensitivity.
- Option 2: Active GPS antenna - Requires more effort for the rework, but provides better GPS sensitivity.

Active GPS antennas have an active RF gain stage, between 3 dB and 25 dB of gain, which results in better GPS sensitivity. Active antennas are powered by L19 at 5 V. Therefore, a 5 V antenna module must be selected if using an active GPS antenna (such as the Taoglas AP.25E.07.0054A GPS Active Patch Antenna) as shown in Table 3-2.

3.1 Option 1: Passive GPS antenna

Table 3-1 Components required for GPS U.FL connector for use with passive GPS antenna

Description	Location (on board)	Vendor	Vendor Part No.	DigiKey Part No.	Mouser Part No.
U.FL connector, 1.25 mm surface mount, 50 Ohm	J11	Molex	Molex 73412-0110	WM5587CT-ND	538-73412-0110
Passive GPS Antenna	Connect to U.FL connector on J11	Multiple	Adafruit 2460 (Passive GPS Antenna uFL – 9 mm x 9 mm -2 dBi gain)	-	485-2460

3.2 Procedure for using a passive GPS antenna

To use a passive GPS antenna, perform the following:

1. Remove R126, which disconnects the on-board GPS antenna.
2. Install the U.FL connector and a zero-ohm resistor in position C216 (a resistor in the position where the schematic shows a capacitor).

NOTE: A solder short is easier to install than a resistor.

3.3 Option 2: active GPS Antenna

Table 3-2 Components required for GPS U.FL connector for use with active GPS antenna

Description	Location (on board)	Vendor	Vendor Part No.	DigiKey Part No.	Mouser Part No.
U.FL connector, 1.25 mm surface mount, 50 Ohm	J11	Molex	Molex 73412-0110	WM5587CT-ND	538-73412-0110
Fixed Inductor, 100nH, 200 mA, 1.5 Ohm, 0402 package	L19	Taiyo Yunden	Taiyo Yunden HK1005R10J-T	587-1531-1-ND	963-HK1005R10J-T
22 pf, 25 V, COG/NPO, 0201 package	C216	Multiple	AVX Corporation 02013A220JAT2A	478-1018-1-ND	581-02013A220J
			AVX Corporation 02013A220JAT4A		
			AVX Corporation 02013A220JAT9A		
			DARFON ELECTRONICS CORP C0201NP0220JFAS		
			DARFON ELECTRONICS CORP C0201NP0220JFTS		
			DARFON ELECTRONICS CORP C0603NP0220JFAS		
			DARFON ELECTRONICS CORP C0603NP0220JFTS		
			MURATA ERIE NORTH AMERICA INC GRM0335C1E220JA01D		
			MURATA ERIE NORTH AMERICA INC GRM0335C1E220JD01D		
			ANASONIC INDUSTRIAL CO ECJZEC1E220J		
			TAIYO YUDEN CO LTD TMK063CG220JT-F		
			TDK CORP OF AMERICA C0603C0G1E220J030BA		
			TDK CORP OF AMERICA C0603C0G1E220JT00NN		
			VENKEL CORP C0201C0G250-220JSNE		
			WALSIN TECHNOLOGY CORP 0201N220J250LT		

Description	Location (on board)	Vendor	Vendor Part No.	DigiKey Part No.	Mouser Part No.
			YAGEO CORPORATION CC0201JRNPO8BN220		
1.0 uF, 6.3 V, X5R, 20%, 0201 package	C217	Multiple	AVX Corporation 02016D105MAT2A	478-7862-1-ND	581-02016D105MAT2A
			MURATA ERIE NORTH AMERICA INC GRM033R60J105MEA2		
			MURATA ERIE NORTH AMERICA INC GRM033R60J105MEA2D		
			MURATA ERIE NORTH AMERICA INC GRM033R60J105MEA2J		
			SAMSUNG ELECTRO-MECHANICS CL03A105MQ3CSNC		
			SAMSUNG ELECTRO-MECHANICS CL03A105MQ3CSNH		
			TAIYO YUDEN CO LTD JMK063ABJ105MP-F		
			TDK CORP OF AMERICA C0603X5R0J105M030BC		
			TDK CORP OF AMERICA C0603X5R0J105MT00NE		
			VENKEL CORP C0201X5R6R3-105MNP		
Active GPS Antenna (5 V antenna module)	Connect to U.FL connector on J11	Multiple	Taoglas AP.25E.07.0054A	931-1141-ND	960-AP25E070054A

3.4 Procedure for using an active GPS antenna

To use an active GPS antenna, perform the following:

1. Remove R126, which disconnects the on-board GPS antenna.
2. Install the U.FL connector *AND* components C216, C217, L19.

NOTE: Many active GPS antennae require more that 1.8V to operate the gain-stage inside the antenna, it may be necessary to stand L19 on its end and use a jumper wire to a higher voltage supply (example 5V) to get the best performance from the selected antenna.

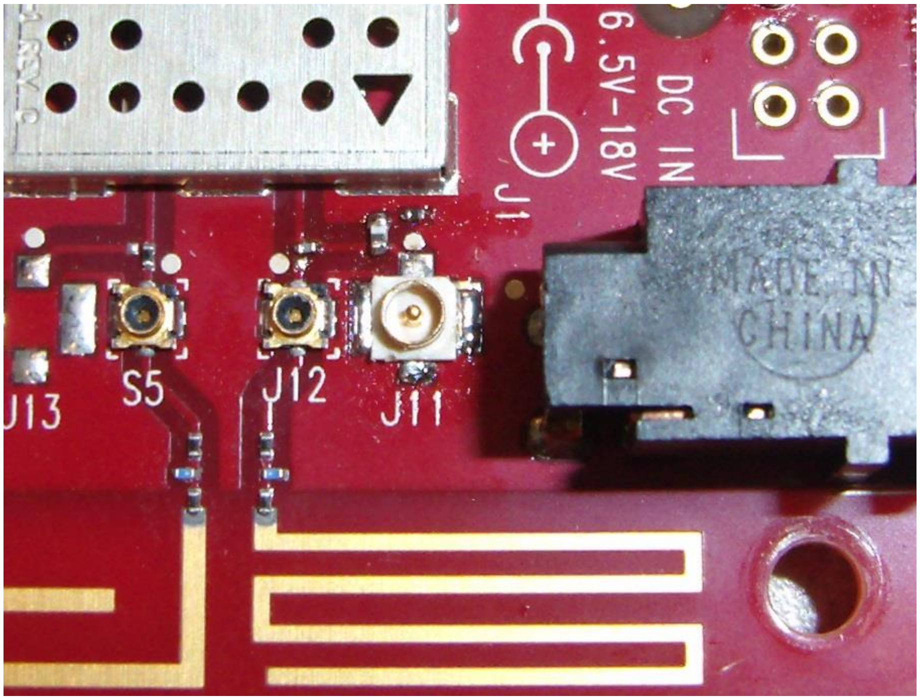

Figure 3-3 U.FL connector rework completed for use with active GPS antenna

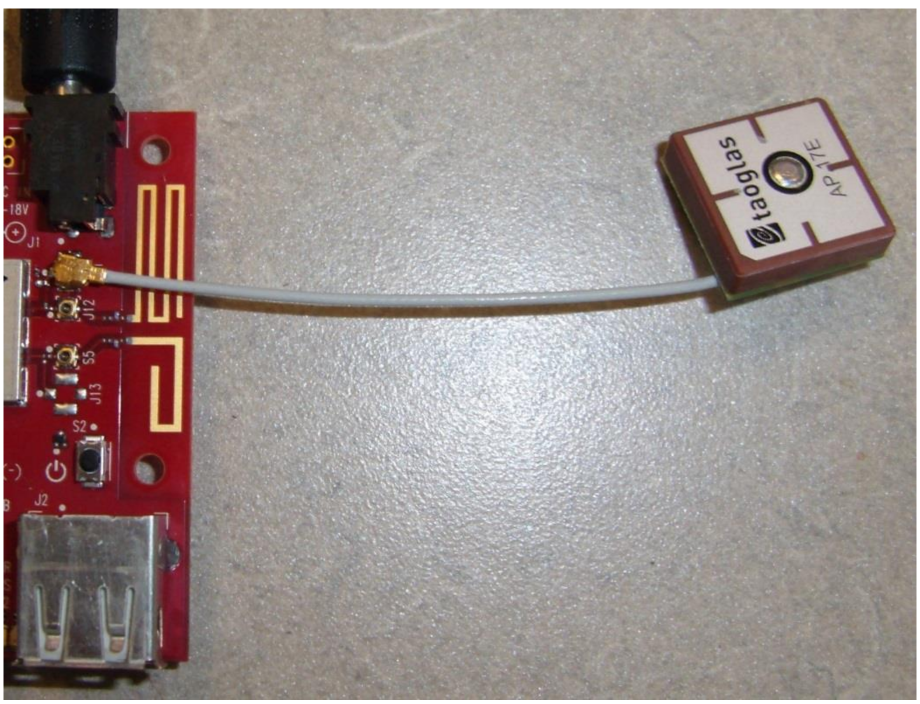

Figure 3-4 Active GPS antenna connected to U.FL connector

3.5 Validate GPS on Android

To validate GPS on Android, perform the following:

1. Flash the Android images.
2. Boot the DragonBoard 410c.
3. Ensure that the adb is enumerated on **Device Manager**.
4. Install the **GPS Test** application.
5. Run the following commands to install GPS Test apk:

```
adb devices
adb root
adb remount
adb install "GPS Test APK"
```

6. Go to **Settings** and change the following options:
   - Location – ON
   - Location mode – High Accuracy or Device only
7. Open the **GPS Test** application from **Application** menu.
   Depending on the outdoor location, antenna position, and weather conditions, the application finds the multiple Space vehicles (SVs) within the reasonable time frame.

   After finding multiple SVs with signal strength average of 35, a 2D/3D fix is achieved.
8. Figure 3-5 shows the 3D Fix (3D-Latitude, Longitude, and Altitude).

NOTE: The time to get First Fix varies on Signal-to-noise ratio (SNR) values and the state of the GPS engine, which runs on the device. Normally, in good-signal condition, maximum accuracy can reach up to 10 feet (3 meters).

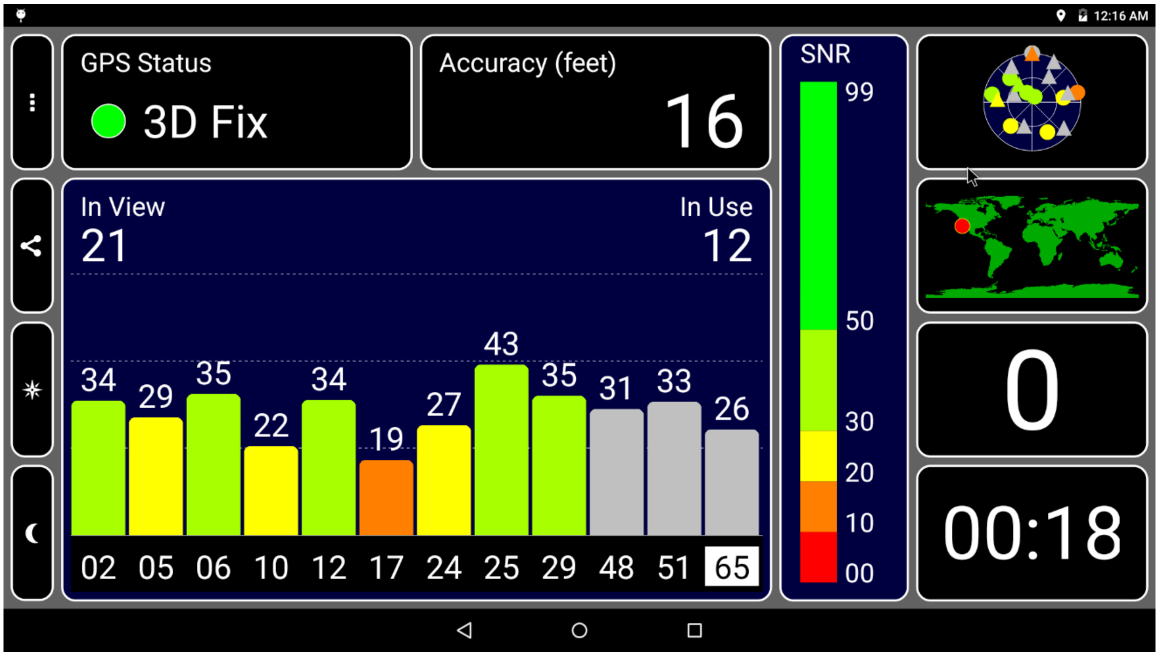

Figure 3-5 3D fix

Figure 3-5 shows 21 in-view satellites (number of satellites that are tracking) and 12 in-use satellites (number of satellites that are used to track).

9. Click **Satellite View** (right side, top corner box in Figure 3-5) to list all the satellites being used and tracked. If it is 3D, then position is obtained (3D – Longitude, Latitude, Altitude). In Figure 3-6, First Fix Time is 3 seconds.

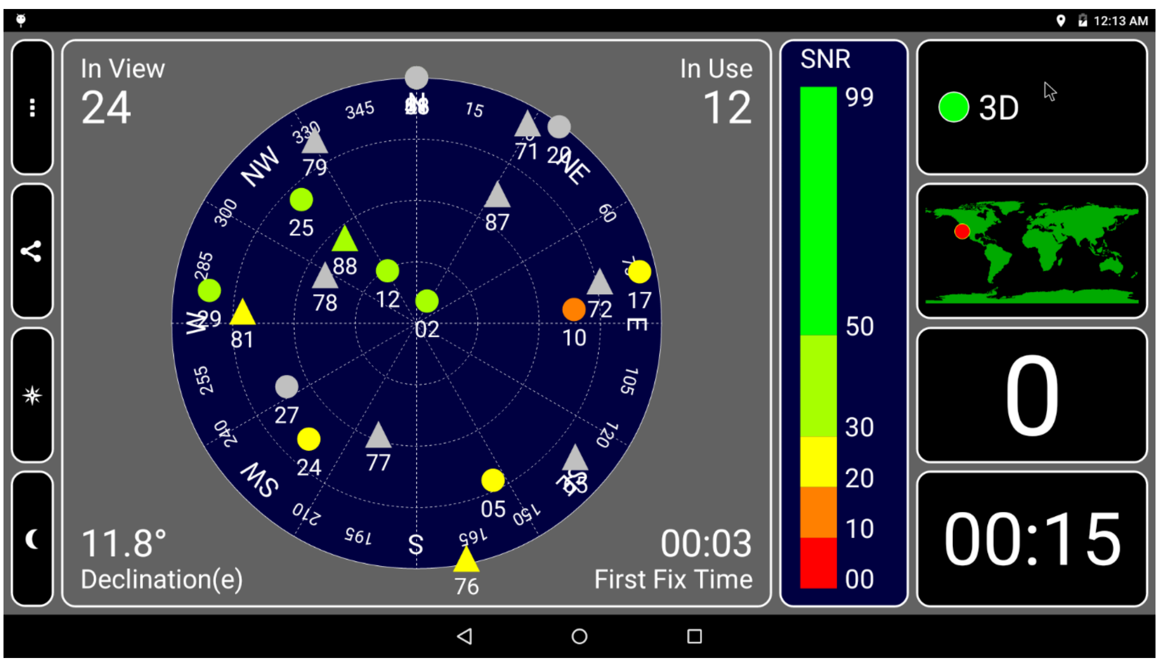

Figure 3-6 Satellite view

10. Click on the *World Map* (right side, second box from the top in Figure 3-6) to show the Longitude and Latitude of the device in WGS84 coordinate system, as shown in Figure 3-7.

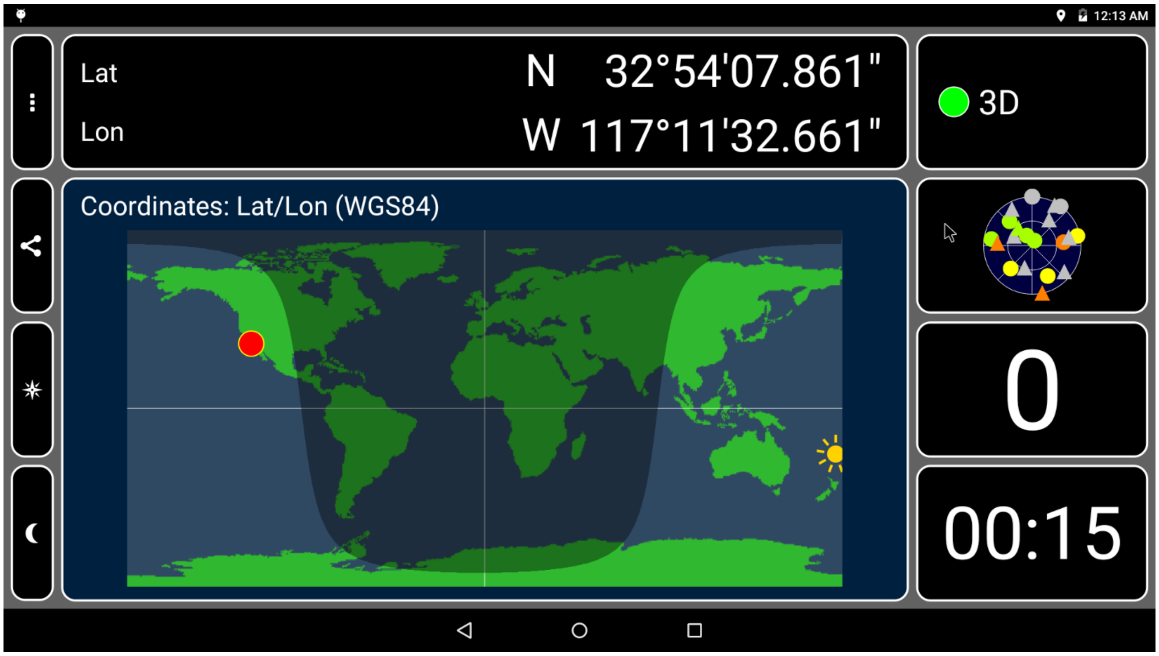

Figure 3-7 Longitude and latitude of the device

11. Click **Speed** (on the right side, third box from the top in Figure 3-7) to show the speed at which the device is moving (in miles per hour) and the Altitude of the device (in feet). Here, the device is in a constant position. So, the speed is 0 mph, as shown in Figure 3-8.

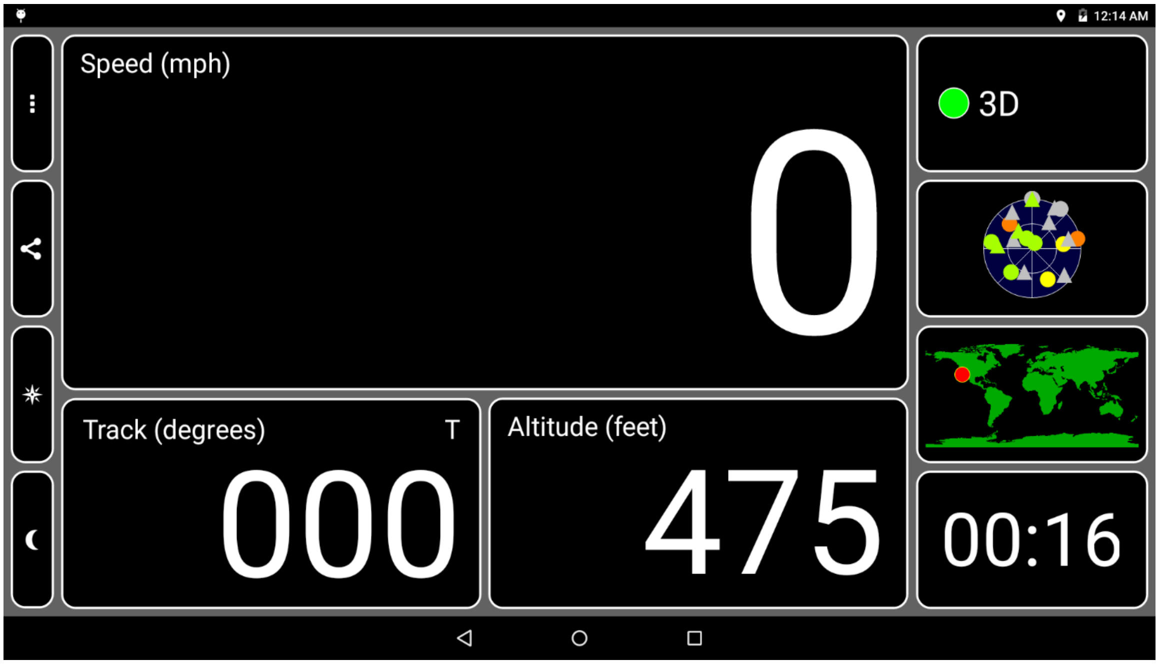

Figure 3-8 Speed

12. Click **Time** (on the right side, fourth box from the top in Figure 3-8) to see the time in UTC format, as shown in Figure 3-9.

Figure 3-9 Time

3.6 Validate GPS on Linux

The GPS software stack runs on the DSP subsystem. The communication between the main CPU and the DSP is done with a specific IPC driver called QRTR. For more information, see *./net/qrtr/* in the kernel source tree. The GPS packages are not installed automatically in the default images. To start the GPS software, some additional packages must be installed. After these packages are installed, the DSP is started automatically (at boot). Any gpsd client can be started and is able to retrieve GPS data.

NOTE: The sensitivity of the onboard antenna is low. Therefore, getting a FIX takes several minutes. Refer to the dedicated application note to install an external antenna for better GPS performance.

To get started with GPS, perform the following:

1. Install the following packages:

```
sudo apt-get install gpsd-clients gnss-gpsd
```

   The package gnss-gpsd brings all the needed dependencies to use the onboard GPS.

2. Use any gpsd client, such as gpsmon or xgps.

EXHIBIT 1

PLEASE READ THIS LICENSE AGREEMENT ("AGREEMENT") CAREFULLY. THIS AGREEMENT IS A BINDING LEGAL AGREEMENT ENTERED INTO BY AND BETWEEN YOU (OR IF YOU ARE ENTERING INTO THIS AGREEMENT ON BEHALF OF AN ENTITY, THEN THE ENTITY THAT YOU REPRESENT) AND Qualcomm Technologies, Inc. ("QTI" "WE" "OUR" OR "US"). THIS IS THE AGREEMENT THAT APPLIES TO YOUR USE OF THE DESIGNATED AND/OR ATTACHED DOCUMENTATION AND ANY UPDATES OR IMPROVEMENTS THEREOF (COLLECTIVELY, "MATERIALS"). BY USING OR COMPLETING THE INSTALLATION OF THE MATERIALS, YOU ARE ACCEPTING THIS AGREEMENT AND YOU AGREE TO BE BOUND BY ITS TERMS AND CONDITIONS. IF YOU DO NOT AGREE TO THESE TERMS, QTI IS UNWILLING TO AND DOES NOT LICENSE THE MATERIALS TO YOU. IF YOU DO NOT AGREE TO THESE TERMS YOU MUST DISCONTINUE AND YOU MAY NOT USE THE MATERIALS OR RETAIN ANY COPIES OF THE MATERIALS. ANY USE OR POSSESSION OF THE MATERIALS BY YOU IS SUBJECT TO THE TERMS AND CONDITIONS SET FORTH IN THIS AGREEMENT.

1.1 **License.** Subject to the terms and conditions of this Agreement, including, without limitation, the restrictions, conditions, limitations and exclusions set forth in this Agreement, Qualcomm® Technologies, Inc. ("QTI") hereby grants to you a nonexclusive, limited license under QTI's copyrights to use the attached Materials; and to reproduce and redistribute a reasonable number of copies of the Materials. You may not use Qualcomm Technologies or its affiliates or subsidiaries name, logo or trademarks; and copyright, trademark, patent and any other notices that appear on the Materials may not be removed or obscured. QTI shall be free to use suggestions, feedback or other information received from You, without obligation of any kind to You. QTI may immediately terminate this Agreement upon your breach. Upon termination of this Agreement, Sections 1.2-4 shall survive.

1.2 **Indemnification.** You agree to indemnify and hold harmless QTI and its officers, directors, employees and successors and assigns against any and all third party claims, demands, causes of action, losses, liabilities, damages, costs and expenses, incurred by QTI (including but not limited to costs of defense, investigation and reasonable attorney's fees) arising out of, resulting from or related to: (i) any breach of this Agreement by You; and (ii) your acts, omissions, products and services. If requested by QTI, You agree to defend QTI in connection with any third party claims, demands, or causes of action resulting from, arising out of or in connection with any of the foregoing.

1.3 **Ownership.** QTI (or its licensors) shall retain title and all ownership rights in and to the Materials and all copies thereof, and nothing herein shall be deemed to grant any right to You under any of QTI's or its affiliates' patents. You shall not subject the Materials to any third party license terms (e.g., open source license terms). You shall not use the Materials for the purpose of identifying or providing evidence to support any potential patent infringement claim against QTI, its affiliates, or any of QTI's or QTI's affiliates' suppliers and/or direct or indirect customers. QTI hereby reserves all rights not expressly granted herein.

1.4 **WARRANTY DISCLAIMER.** YOU EXPRESSLY ACKNOWLEDGE AND AGREE THAT THE USE OF THE MATERIALS IS AT YOUR SOLE RISK. THE MATERIALS AND TECHNICAL SUPPORT, IF ANY, ARE PROVIDED "AS IS" AND WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS OR IMPLIED. QTI ITS LICENSORS AND AFFILIATES MAKE NO WARRANTIES, EXPRESS OR IMPLIED, WITH RESPECT TO THE MATERIALS OR ANY OTHER INFORMATION OR DOCUMENTATION PROVIDED UNDER THIS AGREEMENT, INCLUDING BUT NOT LIMITED TO ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR AGAINST INFRINGEMENT, OR ANY EXPRESS OR IMPLIED WARRANTY ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. NOTHING CONTAINED IN THIS AGREEMENT SHALL BE CONSTRUED AS (I) A WARRANTY OR REPRESENTATION BY QTI, ITS LICENSORS OR AFFILIATES AS TO THE VALIDITY OR SCOPE OF ANY PATENT, COPYRIGHT OR OTHER INTELLECTUAL PROPERTY RIGHT OR (II) A WARRANTY OR REPRESENTATION BY QTI THAT ANY MANUFACTURE OR USE WILL BE FREE FROM INFRINGEMENT OF PATENTS, COPYRIGHTS OR OTHER INTELLECTUAL PROPERTY RIGHTS OF OTHERS, AND IT SHALL BE THE SOLE RESPONSIBILITY OF YOU TO MAKE SUCH DETERMINATION AS IS NECESSARY WITH RESPECT TO THE ACQUISITION OF LICENSES UNDER PATENTS AND OTHER INTELLECTUAL PROPERTY OF THIRD PARTIES.

1.5 **LIMITATION OF LIABILITY.** IN NO EVENT SHALL QTI, QTI'S AFFILIATES OR ITS LICENSORS BE LIABLE TO YOU FOR ANY INCIDENTAL, CONSEQUENTIAL OR SPECIAL DAMAGES, INCLUDING BUT NOT LIMITED TO ANY LOST PROFITS, LOST SAVINGS, OR OTHER INCIDENTAL DAMAGES, ARISING OUT OF THE USE OR INABILITY TO USE, OR THE DELIVERY OR FAILURE TO DELIVER, ANY OF THE MATERIALS, OR ANY BREACH OF ANY OBLIGATION UNDER THIS AGREEMENT, EVEN IF QTI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING LIMITATION OF LIABILITY SHALL REMAIN IN FULL FORCE AND EFFECT REGARDLESS OF WHETHER YOUR REMEDIES HEREUNDER ARE DETERMINED TO HAVE FAILED OF THEIR ESSENTIAL PURPOSE. THE ENTIRE LIABILITY OF QTI, QTI's AFFILIATES AND ITS LICENSORS, AND THE SOLE AND EXCLUSIVE REMEDY OF YOU, FOR ANY CLAIM OR CAUSE OF ACTION ARISING HEREUNDER (WHETHER IN CONTRACT, TORT, OR OTHERWISE) SHALL NOT EXCEED US$10.

2. **COMPLIANCE WITH LAWS; APPLICABLE LAW.** You agree to comply with all applicable local, international and national laws and regulations and with U.S. Export Administration Regulations, as they apply to the subject matter of this Agreement. This Agreement is governed by the laws of the State of California, excluding California's choice of law rules.

3. **CONTRACTING PARTIES.** If the Materials are downloaded on any computer owned by a corporation or other legal entity, then this Agreement is formed by and between QTI and such entity. The individual accepting the terms of this Agreement represents and warrants to QTI that they have the authority to bind such entity to the terms and conditions of this Agreement.

4. **MISCELLANEOUS PROVISIONS.** This Agreement, together with all exhibits attached hereto, which are incorporated herein by this reference, constitutes the entire agreement between QTI and You and supersedes all prior negotiations, representations and agreements between the parties with respect to the subject matter hereof. No addition or modification of this Agreement shall be effective unless made in writing and signed by the respective representatives of QTI and You. The restrictions, limitations, exclusions and conditions set forth in this Agreement shall apply even if QTI or any of its affiliates becomes aware of or fails to act in a manner to address any violation or failure to comply therewith. You hereby acknowledge and agree that the restrictions, limitations, conditions and exclusions imposed in this Agreement on the rights granted in this Agreement are not a derogation of the benefits of such rights. You further acknowledges that, in the absence of such restrictions, limitations, conditions and exclusions, QTI would not have entered into this Agreement with You. Each party shall be responsible for and shall bear its own expenses in connection with this Agreement. If any of the provisions of this Agreement are determined to be invalid, illegal, or otherwise unenforceable, the remaining provisions shall remain in full force and effect. This Agreement is entered into solely in the English language, and if for any reason any other language version is prepared by any party, it shall be solely for convenience and the English version shall govern and control all aspects. If You are located in the province of Quebec, Canada, the following applies: The Parties hereby confirm they have requested this Agreement and all related documents be prepared in English.